

Instructions:

1. Run task_0a.py to generate the vector database if not already generated.
2. Press Run All (or restart kernel and run all cells).
3. You will be prompted to provide input values.

– Task 1: Implement a program which, given (a) a query label, l (b) a user selected feature space, and (c) positive integer k, identifies and visualizes the most relevant k images for the given label l, along with their scores, under the selected feature space.

```
In [ ]: LABEL = input("Provide a valid label name / label ID present in the Caltech101 dataset\n")
FEATURE_SPACE = input("Provide a feature space [color, hog, avgpool, layer3, layer4, layer5]\n")
K = int(input("Enter K, the number of similar images K to find for the selected label\n"))
```

```
In [ ]: # Find the k most similar images for a given query label for the given feature space

from utils.query_input_processor import check_label_is_valid

LABEL = check_label_is_valid(LABEL)
```

Input label: 20
 Downloading dataset if not present.
 Files already downloaded and verified
 Output label: ceiling_fan

```
In [ ]: # Load feature space and representative label vectors for that feature space

from utils.database_utils import retrieve

feature_vectors = retrieve(f'{FEATURE_SPACE}.pt')
rep_label_vectors = retrieve(f'rep_label_{FEATURE_SPACE}.pt')
```

```
In [ ]: from utils.dataset_utils import initialize_dataset
dataset = initialize_dataset()

from utils.distance_utils import get_distance_fn, top_k_min_distance_ranker
top_k = top_k_min_distance_ranker(K, rep_label_vectors[LABEL], feature_vectors)

print("\nK =", K, "most similar images by", FEATURE_SPACE, "feature space using", dataset.feature_space)
```

```
for i in range(len(top_k)):

    distance, image_id, label = top_k[i]

    print(str(i + 1) + ".", "\tImage ID: ", image_id, "\t\tDistance: ", distance)
    display(dataset[image_id][0])
```

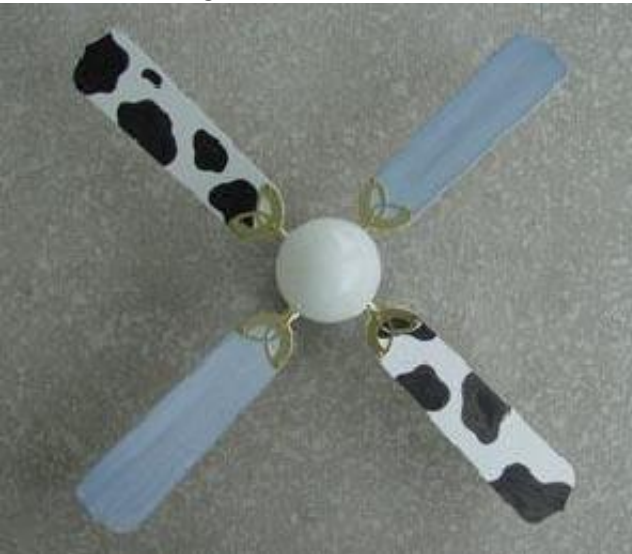
Downloading dataset if not present.
Files already downloaded and verified

K = 10 most similar images by layer3 feature space using distance: cosine

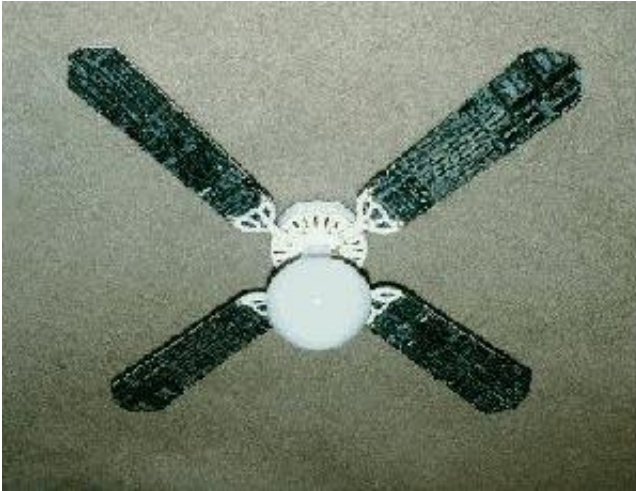
1. Image ID: 3664 Distance: 0



2. Image ID: 3670 Distance: 0.009232580661773682



3. Image ID: 3678 Distance: 0.009331047534942627



4. Image ID: 3666

Distance: 0.014526605606079102



5. Image ID: 3692

Distance: 0.017793655395507812



6. Image ID: 3660

Distance: 0.02272278070449829



7. Image ID: 3694

Distance: 0.022870242595672607



8. Image ID: 3652

Distance: 0.02287733554840088



9. Image ID: 3690

Distance: 0.02668708562850952



10. Image ID: 3650

Distance: 0.026959896087646484

